

Practice Quiz: Perception

Name: __ Date: __

Part A: Multiple Choice

1. Predictive processing says that perception is: A) The passive reception of information from the senses B) The brain's active construction of experience by combining predictions with sensory data C) Entirely determined by external stimuli D) A process that only occurs consciously
2. A prediction error occurs when: A) The brain makes a logical mistake B) The brain's prediction does not match the incoming sensory data C) The agent takes the wrong action D) Two senses give the same information
3. In Bayesian inference, a "prior" is: A) The brain's belief *before* new data arrives B) The most accurate perception possible C) The sensory data from the eyes D) A perfect record of past experience
4. Precision is best described as: A) The accuracy of a measurement B) The brain's confidence in a source of information, determining how much weight it receives C) The resolution of the retina D) The speed of neural processing
5. The Hollow Mask Illusion demonstrates that: A) Our eyes are unreliable B) Strong prior expectations (faces are convex) can override conflicting sensory data C) Hollow objects are always perceived as solid D) The illusion only works with faces
6. In a dark room, your brain relies more on: A) Visual sensory data (because it focuses harder) B) Prior expectations (because visual data is unreliable, so its precision is low) C) Only auditory information D) Memory of the room
7. Change blindness occurs because: A) People aren't paying attention B) The brain predicts continuity and doesn't process every detail of a scene C) The eyes cannot detect changes D) Changes are always too small to perceive

Part B: Short Answer

1. Use the predictive processing framework to explain why you might hear your name called in a noisy crowd, even when nobody said it. What is the prior? What is the sensory data? What is the prediction error?
2. Explain how anxiety might alter perception through precision weighting. Give a specific example — for instance, walking alone in an unfamiliar neighborhood at night.
3. If perception is "the brain's best guess" rather than a direct copy of reality, what implications does this have for eyewitness testimony in criminal trials?